

YUMA, AZ, USA

WMO: 722800

Lat: 32.650N

Lon: 114.600W

Elev: 63

StdP: 100.57

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	5.7	7.1	-15.6	1.0	16.0	-13.2	1.2	17.0	11.0	16.1	9.4	15.5	1.9	30	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	43.9	22.4	42.7	22.3	42.0	22.0	26.3	35.6	25.6	35.1	25.1	34.8	3.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.0	30.4	23.0	17.8	30.8	22.3	17.1	31.2	81.8	35.6	79.1	34.5	76.8	34.5	31.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.6	8.4	7.5	DB	2.2	46.9	1.7	1.3	1.0	47.8	0.0	48.6	-0.9	49.4	-2.1	50.3	(4)
(5)			WB	-1.4	27.8	1.6	1.5	-2.6	28.9	-3.5	29.7	-4.5	30.6	-5.6	31.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	24.6	15.2	16.6	20.1	22.8	26.9	32.0	35.0	32.0	25.5	19.2	14.4	(6)	
(7)		DBStd	7.93	2.78	3.20	3.57	3.71	3.63	3.03	1.94	1.93	2.85	3.58	3.38	2.76	(7)
(8)		HDD10.0	4	2	1	0	0	0	0	0	0	0	0	0	2	(8)
(9)		HDD18.3	354	101	66	24	8	1	0	0	0	0	2	29	125	(9)
(10)		CDD10.0	5336	164	186	312	386	526	659	775	775	659	480	277	138	(10)
(11)		CDD18.3	2644	4	18	78	143	267	409	517	517	409	223	56	3	(11)
(12)		CDH23.3	38961	54	201	894	1686	3539	6339	8535	8473	6153	2544	508	32	(12)
(13)	CDH26.7	25058	3	42	354	816	2066	4241	6074	6010	3980	1314	156	2	(13)	
(14)	Wind	WSAvg	3.2	3.0	2.9	3.0	3.5	3.5	3.9	3.7	3.0	2.8	2.8	3.0	(14)	
(15)	Precipitation	PrecAvg	74	9	6	6	4	1	0	4	16	8	6	11	(15)	
(16)		PrecMax	157	54	31	41	30	9	7	54	87	63	63	42	53	(16)
(17)		PrecMin	1	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	39	11	8	10	8	2	1	10	22	15	13	10	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.9	30.2	34.8	37.8	42.2	45.1	46.0	45.2	42.9	39.0	32.5	26.2	(19)
(20)			MCWB	13.1	14.1	16.7	17.5	20.7	20.5	23.1	22.7	21.8	19.9	16.1	13.9	(20)
(21)		2%	DB	24.8	27.9	32.5	35.7	39.1	42.9	43.9	43.7	41.8	37.1	30.6	23.9	(21)
(22)			MCWB	12.0	13.8	15.8	16.8	18.7	21.1	22.7	22.9	21.5	19.3	15.5	12.5	(22)
(23)		5%	DB	23.0	26.1	30.6	33.2	37.6	41.9	42.6	42.5	40.4	35.0	28.3	22.3	(23)
(24)			MCWB	11.6	13.2	14.8	15.9	18.4	20.5	22.5	23.0	21.5	18.0	15.1	12.2	(24)
(25)		10%	DB	21.8	23.8	28.4	31.4	36.0	40.3	41.5	41.3	38.8	33.0	26.4	20.4	(25)
(26)			MCWB	11.4	12.3	14.2	15.1	17.7	19.9	22.4	23.0	21.7	17.6	14.2	11.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.9	16.9	19.1	19.9	23.5	24.9	27.4	27.3	26.9	23.8	19.6	16.6	(27)
(28)			MCD B	21.9	25.4	32.0	34.6	38.3	39.5	39.4	36.2	33.8	31.5	27.8	22.5	(28)
(29)		2%	WB	14.9	15.5	16.8	17.9	20.7	23.7	25.8	26.4	25.8	22.4	17.6	14.9	(29)
(30)			MCD B	20.2	23.6	29.1	32.6	36.2	37.7	36.9	35.3	33.4	31.3	25.5	21.0	(30)
(31)		5%	WB	13.7	14.5	15.7	16.8	19.3	22.6	25.2	25.8	25.2	21.1	16.3	13.8	(31)
(32)			MCD B	19.5	22.1	27.9	31.1	34.9	36.8	35.7	35.1	33.0	30.1	24.8	19.7	(32)
(33)		10%	WB	12.5	13.5	14.8	15.7	18.2	21.5	24.6	25.3	24.5	19.9	15.2	12.5	(33)
(34)			MCD B	19.0	21.5	26.7	29.4	33.3	36.3	35.6	35.0	33.3	29.4	24.8	18.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.8	12.4	13.6	14.0	14.7	15.1	12.8	12.2	12.5	13.0	12.5	11.3	(35)
(36)			MCDBR	13.7	15.1	16.4	16.5	16.5	16.6	14.3	13.7	14.2	14.9	14.9	13.5	(36)
(37)			MCWBR	6.6	6.6	6.5	6.6	6.3	6.1	4.9	4.4	4.7	5.9	6.5	6.9	(37)
(38)		5% WB	MCD BR	11.0	12.4	14.8	15.3	15.4	14.8	11.9	11.4	11.2	12.0	12.7	11.5	(38)
(39)			MCWBR	6.6	5.9	6.5	6.7	7.2	6.2	4.7	4.5	5.3	5.8	6.9	7.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.290	0.294	0.315	0.324	0.337	0.342	0.386	0.383	0.347	0.322	0.297	0.290	(40)	
(41)		taud	2.495	2.502	2.419	2.403	2.383	2.379	2.370	2.368	2.431	2.454	2.498	2.512	(41)	
(42)		Ebn at Noon	937	968	965	964	947	937	894	895	919	921	921	917	(42)	
(43)		Edh at Noon	86	94	111	118	122	122	122	121	108	98	85	80	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.57	4.56	6.00	7.34	8.15	8.42	7.61	7.10	6.27	5.08	3.87	3.23	(44)	
(45)		RadStd	0.16	0.28	0.19	0.16	0.19	0.29	0.26	0.27	0.20	0.18	0.16	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	-2.17	N/A	-0.44	N/A	N/A	N/A	N/A	N/A		
(47)	Regional (8 neighbors)	+0.33	N/A	-1.61	N/A	-0.36	N/A	N/A	N/A	+129	+92		

Nomenclature: See separate page